Universidad de La Laguna
Instituto de Astrofísica de Canarias

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SPECTRAL PROPERTIES OF POLARIZED SYNCHROTRON AND THERMAL DUST EMISSION FROM POWER SPECTRUM ANALYSES WITH QUIJOTE, WMAP AND PLANCK DATA

Pablo de Ferra Chermaz

Supervisores:

José Alberto Rubiño Martín (IAC)

Resumen

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1 INTRODUCTION

The Cosmic Microwave Background (CMB) provides a unique window to probe the early stages of the Universe and its structure formation. Furthermore, CMB polarization enables to probe the amplitude of primordial Gravitational Waves (GWs; Ade et al., 2021) generated during the inflationary epoch (Kamionkowski, 1999; Kosowsky, 1996). These GWs give rise to the so-called B-mode of CMB polarization. In recent years, many experiments have been observing the CMB anisotropies. For instance, the Wilkinson Microwave Anisotropy Probe (WMAP; Hinshaw et al., 2013), scanning the full sky in five frequency bands between 23 and 94 GHz, and the Planck satellite (Planck Collaboration, 2020) in 9 bands, with polarization measurements in 7 of them, spanning from 30 to 353 GHz. These observations have significantly advanced our understanding of the Universe and motivated further scientific exploration, marking the beginning of a new era of high-precision cosmology with future missions centred in the study of the CMB, such as LiteBIRD (Allys et al., 2022) or other cosmology experiments such as the Square Kilometre Array (SKA; Huynh & Lazio, 2013), offering enhanced sensitivity and resolution.

The primordial B-mode signal might have an amplitude high enough to be within reach of current and upcoming experiments. However, its detection does not depend solely on instrumental sensitivity. A successful discovery critically relies on accurate component separation, since the cosmological signal is contaminated by polarized foreground emission from the Galaxy with significantly higher amplitude (Ade et al., 2021; Planck Collaboration, 2015). Component separation is therefore a central challenge in the design of future CMB experiments such as LiteBIRD (Allys et al., 2022), directly linking the search for primordial B modes to the statistical characterization and astrophysical understanding of polarized emission from the magnetized interstellar medium (ISM).

Microwave observations also contain several astrophysical emission mechanisms other than the CMB, collectively known as foregrounds. Because the primordial B-mode signal has a very low amplitude, the removal of these foregrounds in polarization has become a crucial step for its detection. Two foreground components are known to produce strongly linearly polarized radiation: synchrotron emission, generated by relativistic cosmic-ray electrons spiraling around Galactic magnetic field (GMF) lines and dominating at low frequencies; and thermal dust emission from interstellar dust grains aligned with the magnetic field, which dominates at high frequencies. Both components can exhibit polarization fractions of up to $\sim$20% in certain regions of the sky (Planck Collaboration, 2015).

Anomalous microwave emission (AME), on the other hand, is observed to have a very low polarization fraction (Génova-Santos et al., 2017). If AME originates from spinning dust grains, theoretical models predict negligible polarization levels (Draine et al., 2016; Herman, D. et al., 2023). Another relevant foreground is free-free emission, produced by free electrons interacting with ionized gas; however, this mechanism generates radiation that is effectively unpolarized (Gold et al., 2011).

Synchrotron emission accounts for most of the radio emission from Active Galactic Nuclei (AGNs), which are thought to be powered by supermassive black holes in galaxies and quasars (Blandford & Znajek, 1977; Ishibashi, W. & Courvoisier, T. J.-L., 2011). This emission also dominates the radio continuum from star-forming galaxies, such as our own, at frequencies below $\nu \sim 30$ GHz. The relativistic electrons in nearly all synchrotron sources typically exhibit power-law energy distributions. The energy flux distribution for synchrotron radiation is given by $S_\nu \propto \nu^\alpha$, or equivalently, in terms of the brightness temperature as $T_\nu \propto \nu^\beta$, where $\beta = \alpha - 2$ (Rybicki & Lightman, 1979). This frequency dependence of the brightness temperature will be referred to as the spectral index. The spectral index depends on the energy-independent spectral energy slope s of the cosmic-ray electron spectrum, which is typically assumed to have a value of $s = -3$ in the frequency range from 10 to 30 GHz, but varying at lower frequencies (Padovani et al., 2021). Spatial variations of β reflect spatial variations of the cosmic-ray electrons and magnetic field properties in the interstellar medium along the line of sight. Full-sky maps of Galactic synchrotron emission have already been derived in Krachmalnicoff et al. (2018) or Rubiño-Martín et al. (2023), and their analyses clearly show spatial variations of the spectral index, as discussed below.

At higher frequencies, the dominant polarized foreground is thermal emission from interstellar dust grains aligned with the GMF. This emission dominates the microwave sky above approximately 100 GHz and is partially linearly polarized. The frequency spectrum of thermal dust emission is well described by a modified black body (grey body), $I_d(\nu) \propto \nu^{\beta_d} B_\nu(T_d)$, where $B_\nu(T_d)$ is the Planck black-body spectrum at dust temperature T_d. Using combined Planck and WMAP data, Planck Collaboration et al. (2015) determined the dust spectral index in both intensity and polarization, finding $\beta_d \simeq 1.59$ for $T_d \simeq 19.6\,\mathrm{K}$, while **?** derived a mean polarization spectral index of $\beta_d = 1.53 \pm 0.02$ from PR3 maps at 353 GHz, with the dust EE and BB power spectra following power laws in multipole and exhibiting a significant E/B asymmetry. Since both dust and synchrotron polarization are aligned by the same GMF, a spatial cross-correlation between the two components is expected and has indeed been detected (Choi & Page, 2015; **?**).

The sensitivities of WMAP and Planck alone do not allow a sufficiently accurate characterization of the polarized synchrotron signal at intermediate and high Galactic latitudes, leaving important uncertainties in the modelling of this foreground component. This is significantly improved by the QUIJOTE experiment (Rubiño-Martín et al., 2023), which observes the polarized Northern sky in the 10–40 GHz range, precisely where synchrotron emission is strongest and where Faraday rotation effects are not expected to be significant (Fuskeland et al., 2021). Combining QUIJOTE with WMAP and Planck data, de la Hoz et al. (2023) performed one of the first reconstructions of the polarized synchrotron spectral index in real space, finding statistically significant spatial variations across the sky, with an average value of $\beta = -3.08$ and a dispersion of 0.13. More recently, Casas et al. (2025) studied the spatial variability of polarized synchrotron emission using QUIJOTE MFI data alone, reinforcing the evidence for spatial variability in polarized synchrotron emission and highlighting the importance of combining QUIJOTE observations with complementary data from WMAP, Planck, and future low-frequency surveys in order to reduce noise contamination and improve the robustness of spectral-index estimates.

In this work, we characterize the spectral properties of polarized synchrotron and thermal dust emission through power spectrum analyses of the combined QUIJOTE MFI DR1 (11, 13, 17, and 19 GHz), WMAP 9-year (K, Ka, Q, V, and W bands), and Planck PR3 (LFI and HFI, 28.4–353 GHz) datasets. Both foreground components are modelled simultaneously by fitting a parametric SED model to the EE and BB angular power spectra (APS) and their cross-spectra across this wide frequency range. A key aspect of this analysis is the explicit treatment of the spatial cross-correlation between polarized synchrotron and thermal dust emission. Their cross-spectra encode complementary information about the geometry and coherence of the Galactic magnetic field along the line of sight (Planck Collaboration et al., 2015; Choi & Page, 2015). The analysis is carried out over the full Northern sky footprint of QUIJOTE, using three Galactic latitude masks to assess the sensitivity of the results to Galactic plane contamination. The resulting foreground characterization is intended to serve as legacy input for component separation in upcoming CMB polarization experiments such as LiteBIRD (Allys et al., 2022).

2 OBJECTIVES

The main objective of this work is to characterize the spectral properties of the two dominant polarized Galactic foregrounds: synchrotron and thermal dust emission, using harmonic-space (power spectrum) tools. The analysis is performed on the combined dataset of QUIJOTE MFI DR1 maps at 11 and 17 GHz (Rubiño-Martín et al., 2023), all five WMAP 9-year frequency bands (K, Ka, Q, V, and W, Bennett et al., 2013), and the three LFI (Planck Collaboration, 2016a) and four HFI polarization bands (Planck Collaboration et al., 2020a) from Planck PR3, spanning a total frequency range from 11 to 353 GHz. This broad frequency coverage enables us to simultaneously fit a parametric spectral energy distribution (SED) model for both foreground components.

A second key objective is the characterization of the spatial cross-correlation between polarized synchrotron and thermal dust emission. Since both foregrounds are polarized by the same GMF, their EE and BB angular power spectra are expected to be partially correlated. Measuring this cross-correlation, alongside the auto-spectra of each component, provides complementary constraints on the geometry and coherence of the GMF and improves the accuracy of foreground separation in the CMB polarization context (Choi & Page, 2015).

Our results will be compared to prior power-spectrum studies that did not include QUIJOTE data. Martire et al. (2022) characterized the polarized synchrotron emission using WMAP and Planck data alone, obtaining a spectral index of $\beta_s = -3.00 \pm 0.10$ and a ratio of B- to E-mode amplitudes of $BB/EE \simeq 0.2$. Krachmalnicoff et al. (2018) obtained a steeper mean value of $\beta_s = -3.22 \pm 0.08$ when including the lower-frequency S-PASS data at 2.3 GHz.

For the thermal dust component, our results will be compared with those of Planck Collaboration et al. (2020b), who used Planck PR3 maps at 353 GHz to characterise dust polarization power spectra over sky regions covering 24 to 71% of the sky. They found power-law multipole indices $\alpha_{EE} = -2.42 \pm 0.02$ and $\alpha_{BB} = -2.54 \pm 0.02$, a significant E/B power asymmetry, and variations of the spectral exponents across sky regions. The dust polarization SED was well described by a modified black body with spectral index $\beta_d = 1.53 \pm 0.02$, with no evidence for frequency decorrelation between 100 and 353 GHz. For the synchrotron-dust cross-correlation, the benchmark is provided by Choi & Page (2015), who showed that a joint foreground model including the synchrotron-dust cross-correlation fits all WMAP and Planck frequency cross-spectra. These results constitute the main observational benchmarks against which our joint synchrotron-plus-dust analysis will be validated.

Beyond the immediate scientific goals, this work aims to provide robust foreground constraints and data products, including spectral index estimates and validated foreground models, that will serve as essential inputs for component separation in forthcoming CMB polarization experiments, including (Allys et al., 2022) and other ground-based and space facilities.

The analysis is carried out over the full sky footprint available to the experiments, using three Galactic masks with different latitude cuts to assess the sensitivity of our results to Galactic plane emission.

3 DATA

All data used in this work consist of three independent maps for each frequency band: one intensity map, corresponding to the Stokes parameter I, and two maps describing linear polarization, given by the Stokes parameters Q and U. In this work, we consider polarization data from three experiments: QUIJOTE-MFI, WMAP, and Planck, which are described below.

3.1. Q-U-I JOint TEnerife (QUIJOTE)

The QUIJOTE experiment (Rubiño-Martín et al., 2023) observes the visible sky from the Teide Observatory (latitude $+28.3°$), covering all regions with elevations greater than 30° in both intensity and polarization. The experiment consists of two telescopes equipped with three polarimetric instruments: the Multifrequency Instrument (MFI), operating in the 10–20 GHz range; the Thirty-GHz Instrument (TGI); and the Forty-GHz Instrument (FGI).

In this work we use data from the 11 and 17 GHz bands of the MFI, which have angular resolutions of 55 and 38 arcmin, respectively. The data are obtained from the Legacy Archive for Microwave Background Data Analysis (LAMBDA)[1]. The 11 GHz band is particularly useful for characterizing the polarization of low-frequency foregrounds, primarily synchrotron emission (Adak et al., 2025; Casas et al., 2025) and Anomalous Microwave Emission (AME, Fernández-Torreiro et al., 2023; González-González et al., 2025).

[1] https://lambda.gsfc.nasa.gov/product/quijote/index.html

3.2. Wilkinson Microwave Anisotropy Probe (WMAP)

WMAP was a space-based experiment that observed the total intensity and polarization of the full sky across five frequency bands, spanning the frequency range from 23 to 94 GHz. In this work, we make use of all five frequency bands from the 9-year data release (Bennett et al., 2013): the K-band (23 GHz, $FWHM$ = 52.8 arcmin), Ka-band (33 GHz, $FWHM$ = 39.6 arcmin), Q-band (41 GHz, $FWHM$ = 31.2 arcmin), V-band (61 GHz, $FWHM$ = 21.0 arcmin), and W-band (94 GHz, $FWHM$ = 13.2 arcmin). All maps are downloaded from the LAMBDA database[2].

3.3. Planck

Planck was a space-based experiment operating two complementary instruments: the Low Frequency Instrument (LFI) and the High Frequency Instrument (HFI), together observing the full sky in intensity and polarization across nine frequency channels, from 30 to 857 GHz. In this work, we use data from the third public release (PR3) of the Planck mission, available through the Planck Legacy Archive[3] (PLA).

3.3.1 Low Frequency Instrument (LFI)

The LFI instrument covered three frequency bands at 28.4, 44.1, and 70.4 GHz. We use the PR3 frequency maps from all three LFI channels (Planck Collaboration, 2016a). The original angular resolutions correspond to effective beams of 32.65, 27.94, and 13.01 arcmin at 28.4, 44.1, and 70.4 GHz, respectively. All maps are downgraded to a HEALPix[4] resolution of $N_{\rm side} = 512$ to maintain a common pixelization with the QUIJOTE and WMAP datasets.

3.3.2 High Frequency Instrument (HFI)

The HFI instrument provides six frequency bands extending up to 857 GHz. In this analysis we use the PR3 frequency maps from the HFI channels up to 353 GHz (100, 143, 217, and 353 GHz) (Planck Collaboration et al., 2020a). These four channels provide polarization information, including Stokes Q and U maps, and are therefore used in our polarization analysis. The higher-frequency HFI channels at 545 and 857 GHz are not included, as they do not provide polarization measurements. To ensure a consistent pixelization across the QUIJOTE, WMAP, and Planck datasets, all selected maps are downgraded to $N_{\rm side} = 512$.

4 METHODOLOGY

In this section we describe the methodology followed to characterise Galactic foreground emission using multi-frequency angular power spectrum analysis. The approach relies entirely on harmonic-space statistics, avoiding the complications of real-space fitting, and exploits the broad frequency coverage provided by the combination of QUIJOTE, WMAP, and Planck datasets. We first introduce the mathematical framework of angular power spectrum analysis (Section 4.1.), then describe the construction of the analysis masks (Section 4.2.), the data processing steps (beam in Section 4.3.1, unit corrections in Section 4.3.2, noise in Section 4.3.3, and color corrections in Section 4.3.4), the simulations used to estimate uncertainties (Section 4.4.), the foreground models (Section 4.5.), and finally, the Bayesian parameter inference (Section 4.6.).

4.1. Angular power spectrum (APS) analysis

The APS is a statistical tool that describes the distribution of signal power as a function of angular scale on the sky. It is a particularly powerful instrument for studying the statistical properties of diffuse emission, since it allows one to separate contributions from different physical processes through their characteristic frequency and multipole dependencies. Rather than working directly

[2] https://lambda.gsfc.nasa.gov/product/wmap/current/index.html
[3] https://pla.esac.esa.int/
[4] https://healpy.readthedocs.io/en/latest/

in pixel space, all analyses presented in this work are carried out in harmonic space, where the APS constitutes the fundamental observable.

The APS is expressed in terms of the spherical harmonic coefficients $a_{\ell m}$, which encode the sky signal at each multipole ℓ and azimuthal order m. The multipole ℓ is inversely proportional to the angular scale $\theta \sim 180^\circ/\ell$, so that large angular scales correspond to low-ℓ values, and small angular scales to high-ℓ values. For reference, $\ell = 200$ corresponds to approximately 1°, which is comparable to the angular resolution of the QUIJOTE-MFI instrument. The cross-power spectrum between two frequency maps X and Y is defined as:

$$C_\ell^{X\times Y} = \left\langle \frac{\sum_m a_{\ell m}^X (a_{\ell m}^Y)^*}{2\ell+1} \right\rangle, \tag{1}$$

where X and Y denote the two frequency bands. When $X = Y$ this reduces to the auto-power spectrum. Cross-spectra probe the signal component common to two independent observations, suppressing the contribution of uncorrelated noise. By combining auto- and cross-spectra from 14 frequency bands (2 QUIJOTE, 5 WMAP and 7 Planck channels), we form a total of 105 spectra (14 auto and 91 cross), greatly improving the constraining power on the foreground model parameters.

All power spectra in this work are computed using the NaMaster Python library[5] (Alonso et al., 2019), which implements the pseudo-C_ℓ method with an analytical treatment of the mode-coupling matrix induced by the sky mask. In particular, the E- and B-mode polarization fields are treated with B-mode purification (Smith, 2006), which suppresses E-to-B leakage arising from the application of a partial sky mask. The power spectra are computed in nine multipole bins of width $\Delta\ell = 20$ over the range $20 \le \ell \le 199$, with bin edges defined as

$$\ell_{\rm bin}^{(k)} \in [\ell_1^{(k)}, \ell_2^{(k)}] = [20 + 20(k-1), 39 + 20(k-1)], \quad k = 1, \dots, 9. \tag{2}$$

The effective multipole $\ell_{\rm eff}$ reported for each bin is computed by NaMaster as the weighted average of the individual multipoles within the bin:

$$\ell_{\rm eff}^{(k)} = \frac{\sum_{\ell \in {\rm bin\,k}} w_\ell \, \ell}{\sum_{\ell \in {\rm bin\,k}} w_\ell}, \tag{3}$$

where w_ℓ are the per-multipole weights assigned to the bandpower. In this work uniform weights are used, so $\ell_{\rm eff}^{(k)}$ reduces to the arithmetic mean of all integer multipoles in the k-th bin, yielding $\ell_{\rm eff} = 29, 49, 69, \dots, 189$ for the nine bins.

4.2. Masks

The analysis is restricted to the sky region accessible to the QUIJOTE-MFI telescope (Section 3.1.), which operates from the Teide Observatory and therefore covers a declination band of approximately $-30^\circ < \delta < +90^\circ$. To mitigate contamination from the bright Galactic plane, three analysis masks are constructed by combining the QUIJOTE low-declination survey footprint (Rubiño-Martín et al., 2023) with Galactic latitude cuts at $|b| > 10^\circ$, $|b| > 15^\circ$, and $|b| > 20^\circ$. Each binary mask is apodized using the C2 scheme of NaMaster with an apodization scale of 5°, which tapers the mask smoothly to zero at its edges and suppresses ringing effects in harmonic space. The three masks are shown in Figure 1. The effective sky fractions after apodization are approximately $f_{\rm sky} \approx 0.30$, 0.26, and 0.23 for the 10, 15, and 20 degree cuts, respectively. The masks are defined on a HEALPix grid at $N_{\rm side} = 512$ and are identical for all frequency bands.

[5] https://namaster.readthedocs.io/en/latest/

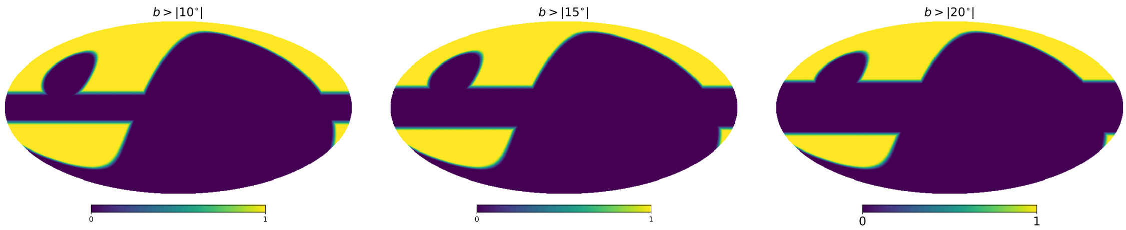

Figure 1: Analysis masks combining the QUIJOTE low-declination survey footprint with Galactic latitude cuts of $|b| > 10°$ (left), $|b| > 15°$ (middle), and $|b| > 20°$ (right), after C2 apodisation with a 5° border.

4.3. Data corrections

The raw power spectra computed from the observed maps contain systematic contributions from instrumental effects that must be accounted for before a physical interpretation can be drawn. In the ideal case, the measured C_ℓ coefficients would be proportional to the square of the theoretical spherical harmonic coefficients, $|a_{\ell m}^{\rm theo}|^2$. In practice, several multiplicative and additive corrections are applied, as described below.

4.3.1 Beam and pixel window functions

The finite angular resolution of each instrument and the pixelization of the HEALPix maps (Górski et al., 2005) introduce multiplicative window functions that suppress power at high multipoles. The pixel window function, w_ℓ, arises from the averaging of the sky signal over each pixel and depends solely on $N_{\rm side}$; for $N_{\rm side} = 512$ this correction is small but non-negligible at $\ell > 500$. The instrumental beam is characterised by the window function B_ℓ, which captures the effective angular resolution of the experiment and causes an exponential suppression of power above the beam scale.

For each frequency band, the beam window function B_ℓ is loaded from the official data products of each experiment and interpolated to the effective multipoles of the binning scheme. For QUIJOTE-MFI, the beam is provided as a tabulated array per frequency channel in a FITS file. For WMAP, it is provided as a normalised ASCII table. For Planck LFI and HFI bands, the beam window functions are taken from the respective official FITS products, with separate T, E, and B transfer functions for HFI channels.

Both cases — auto- and cross-spectra — are handled by a single correction formula. Defining the effective transfer function for band i as $\mathcal{W}_\ell^i \equiv w_\ell\, B_\ell^i$, where w_ℓ is the HEALPix pixel window function (common to all bands at fixed $N_{\rm side}$) and B_ℓ^i is the instrumental beam window function of band i, the corrected spectrum between bands i and j is:

$$C_\ell^{i\times j,\,\rm corr} = \frac{C_\ell^{i\times j,\,\rm sky} - \delta_{ij}\, N_\ell^i}{\mathcal{W}_\ell^i\, \mathcal{W}_\ell^j}, \tag{4}$$

where δ_{ij} is the Kronecker delta, N_ℓ^i is the noise power spectrum of band i (see Section 4.3.3), and the denominator deconvolves the combined effect of the pixel window and the beam of each band. The Kronecker delta ensures that the noise bias is subtracted only for auto-spectra ($i = j$); for cross-spectra between different frequency bands the noise contributions of the two detectors are statistically independent and thus average to zero in the cross-correlation, so no noise subtraction is required.

4.3.2 Unit conversion

All sky maps are converted to a common unit of $\rm K_{RJ}$ (Rayleigh–Jeans brightness temperature) before computing the power spectra. The unit conversion factor from thermodynamic CMB temperature $\rm K_{CMB}$ to $\rm K_{RJ}$ is:

$$u(\nu) = \frac{c^2}{2k_B\nu^2}\left.\frac{\partial B_\nu}{\partial T}\right|_{T_{\rm CMB}}, \tag{5}$$

where B_ν is the Planck function, k_B is the Boltzmann constant, c the speed of light, and $T_{\rm CMB} = 2.725$ K. For Planck HFI bands (100–353 GHz), the conversion uses the official bandpass-averaged unit conversion coefficients from the Planck 2018 data release (Planck Collaboration et al., 2020a), accounting for the instrumental bandpass shape. For all remaining bands the conversion is evaluated analytically at the effective band-centre frequency. The resulting unit conversion factor applied to a power spectrum of the pair (i, j) is $u(\nu_i) \times u(\nu_j)$, where $i = j$ for auto-spectra and $i \neq j$ for cross-spectra.

4.3.3 Noise power spectrum

The instrumental noise contributes an additive bias to the auto-power spectrum that must be estimated and removed (Equation 4). The noise estimation strategy is specific to each experiment.

For QUIJOTE-MFI and Planck, the noise power spectrum for each auto-spectrum is estimated from the half mission difference map (HMDM). The HMDM is constructed from two independent half-mission data subsets, h_1 and h_2, and their associated weight maps, w_1 and w_2, as (Rubiño-Martín et al., 2023):

$$n = \frac{h_1 - h_2}{\sqrt{(w_1 + w_2)(w_1^{-1} + w_2^{-1})}}, \tag{6}$$

where $w_i = 1/\sigma_i^2$. This construction ensures that the difference map retains the noise level of the combined weighted-sum map while containing no sky signal (Planck and AMI Collaborations et al., 2013; Planck Collaboration et al., 2016b). For WMAP, no half-mission maps are available; the noise power spectrum is instead estimated as the mean over 100 noise realisations (see Section 4.4.2):

$$N_\ell = \left\langle C_\ell^{\rm noise,\, sim} \right\rangle_{100\,\rm sims}. \tag{7}$$

4.3.4 color corrections

color corrections account for the mismatch between the spectral shape of the foreground emission and the spectral response of the instrument bandpass. Following the procedure described in Planck Collaboration et al. (2020c), color correction factors are evaluated as a function of the foreground spectral index and incorporated in the MCMC fitting stage. The correction coefficients for each band and component (synchrotron and thermal dust) are computed using the FASTCC PYTHON and IDL code[6] (Peel et al., 2022), which evaluates the bandpass-weighted ratio between the assumed foreground SED and a flat (Rayleigh–Jeans) reference spectrum.

The color correction factors are represented as second-order polynomials in the relevant spectral index α:

$$\mathrm{CC}(\alpha) = a_0 + a_1\,\alpha + a_2\,\alpha^2, \tag{8}$$

where $\alpha = \beta_s + 2$ for synchrotron and $\alpha = \beta_d + 2$ for dust in Rayleigh-Jeans units. These polynomial coefficients are pre-computed using FASTCC and stored in a FITS file loaded at runtime. Corrections are applied in a self-consistent iterative procedure: a first MCMC pass is performed without color corrections to obtain initial spectral index estimates; these are then used to evaluate the correction factors, which are incorporated in a second MCMC pass. In the spectral model, the color correction enters as a divisive factor for each band:

$$C_\ell^{\rm sync,CC}(\nu_i, \nu_j) = \frac{A_s\,(\ell/\ell_{\rm ref})^{\alpha_s}\,(\nu_i/\nu_{\rm ref})^{\beta_s}\,(\nu_j/\nu_{\rm ref})^{\beta_s}}{\mathrm{CC}_s(\nu_i)\,\mathrm{CC}_s(\nu_j)}, \tag{9}$$

and analogously for the dust and synchrotron-dust cross-correlation terms.

[6] https://github.com/mpeel/fastcc

4.4. Simulations

The statistical uncertainties on the measured power spectra are estimated from Monte Carlo simulations. The error budget for the auto-spectrum has two independent contributions: the variance of the sky-signal spectra across realisations, and the variance of the noise spectra. These are estimated from two independent sets of $N_{\rm sim} = 100$ simulations, and the total uncertainty is obtained as their quadrature sum:

$$\sigma_\ell = \sqrt{\sigma_\ell^{2,\,\rm sky+noise} + \sigma_\ell^{2,\,\rm noise}}. \tag{10}$$

For cross-spectra, where no noise correction is applied, only the sky-plus-noise simulation variance is used as the error estimate.

4.4.1 Sky signal simulations

Realistic simulated polarisation sky maps for each frequency band and experiment are generated using the Python Sky Model (PySM) package[7] (Thorne et al., 2017). PySM is a widely used Python library for sky emission modelling in CMB analyses (e.g. Planck Collaboration et al. 2020c; Martire et al. 2022).

The simulation includes both polarised and intensity foreground components. The polarised foreground components are: synchrotron emission modelled as a power law with a spatially constant spectral index (**s1**); thermal dust emission described as a single-component modified blackbody (**d1**); and a lensed CMB polarisation realisation (**c1**), computed using the Taylens code (Næss & Louis, 2013) with unlensed C_ℓ produced by CAMB[8]. Intensity foreground components are also included: spinning dust (AME) modelled as a sum of two spinning dust populations (**a1**; Planck Collaboration 2016b), and free-free emission following the analytic Commander model (**f1**).

Simulations are produced independently for all 14 frequency bands. The sky emission at each band-centre frequency is obtained from PySM, converted to thermodynamic CMB temperature units (μK$_{\rm CMB}$), and subsequently convolved in harmonic space with the instrumental beam window function of the corresponding experiment. Each simulated map is stored in HEALPix format at $N_{\rm side} = 512$, in units of mK for QUIJOTE and WMAP and K for Planck.

4.4.2 Noise simulations

Noise simulations are produced independently for each frequency band and experiment, reflecting the distinct noise properties of each instrument.

QUIJOTE-MFI. The noise in the QUIJOTE-MFI maps is spatially correlated and anisotropic, and the two channels within each MFI horn (e.g. 11 and 17 GHz) exhibit significant noise correlations both in intensity ($\sim$ 60–80%) and in polarisation ($\sim$ 20%; Rubiño-Martín et al. 2023). The QUIJOTE noise simulations used in this work are those described in Rubiño-Martín et al. (2023), which account for the measured anisotropy, spatial correlations, and inter-channel covariance structure. These simulations are also used to construct the block-diagonal covariance matrix for the MCMC likelihood (see Section 4.6.).

WMAP. WMAP data are not affected by $1/f$ noise, so the noise is well described by a per-pixel Gaussian model. Following Bennett et al. (2013), $N_{\rm sim} = 100$ noise realisation maps are generated as independent Gaussian random fields with zero mean and pixel-dependent standard deviation derived from the per-pixel noise sensitivity σ_0 and the effective number of observations $N_{\rm obs}$:

$$\sigma_{\rm pix} = \frac{\sigma_0}{\sqrt{N_{\rm obs}}}. \tag{11}$$

The Q and U polarisation components are generated independently with their respective per-pixel noise levels. The mean of the 100 noise simulation spectra is used as the noise bias subtracted from the WMAP auto-spectra.

[7] https://pysm3.readthedocs.io/en/latest/
[8] https://www.camb.info/

Planck. For Planck LFI and HFI bands, $N_{\rm sim} = 100$ per-band noise simulations are downloaded from the Planck Legacy Archive (Planck Collaboration et al., 2020c; Aghanim et al., 2020). These simulations are generated using an MCMC-based noise estimation procedure (Planck and AMI Collaborations et al., 2013) that characterises the noise power spectrum including contributions from the $1/f$ noise component of the detectors, and each simulation map is processed identically to the data in the analysis pipeline.

4.5. Foreground emission model

The measured power spectra over the frequency range 11–353 GHz are dominated by two Galactic emission components in polarisation: synchrotron radiation and thermal dust emission. A cross-correlation term between the two components is included to account for any spatial correlation along the line of sight. The total model power spectrum between frequencies ν_i and ν_j is:

$$C_\ell(\nu_i,\nu_j) = C_\ell^{\rm sync}(\nu_i,\nu_j) + C_\ell^{\rm dust}(\nu_i,\nu_j) + C_\ell^{\rm corr}(\nu_i,\nu_j). \tag{12}$$

4.5.1 Synchrotron emission

Synchrotron radiation is the dominant polarised foreground at frequencies below ∼70 GHz. Its angular power spectrum is well described by a power law in ℓ:

$$C_\ell^{\rm sync}(\nu_i,\nu_j) = A_s \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_s} \left(\frac{\nu_i}{\nu_{\rm ref}}\right)^{\beta_s} \left(\frac{\nu_j}{\nu_{\rm ref}}\right)^{\beta_s}, \tag{13}$$

where A_s is the synchrotron amplitude at the reference multipole $\ell_{\rm ref} = 80$ and reference frequency $\nu_{\rm ref} = 23.0$ GHz; α_s is the angular power spectral index; and β_s is the frequency spectral index in Rayleigh-Jeans units. The EE and BB modes are fitted separately.

4.5.2 Thermal dust emission

Thermal dust emission dominates polarised foreground emission above ∼70 GHz, with a SED following a modified blackbody (MBB) law. In Rayleigh-Jeans brightness temperature units, the MBB scaling relative to the dust reference frequency $\nu_0 = 353$ GHz is:

$$S_d(\nu) = \left(\frac{\nu}{\nu_0}\right)^{\beta_d} \frac{B_\nu(\nu,T_d)}{B_\nu(\nu_0,T_d)}, \tag{14}$$

where B_ν is the Planck function and $T_d = 19.6$ K is the dust temperature, fixed in accordance with Planck measurements (Planck Collaboration, 2016b). The dust model power spectrum is:

$$C_\ell^{\rm dust}(\nu_i,\nu_j) = A_d \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_d} S_d(\nu_i)\, S_d(\nu_j), \tag{15}$$

where A_d is the dust amplitude at $\ell_{\rm ref} = 80$ and $\nu_0 = 353$ GHz; α_d is the angular power spectral index; and β_d is the MBB frequency spectral index.

4.5.3 Synchrotron-dust cross-correlation

A non-zero spatial correlation between synchrotron and thermal dust emission can arise when the Galactic magnetic field aligns the polarisation angle of both components along the line of sight (Choi & Page, 2015). Following Planck Collaboration et al. (2016a), the cross-correlation term is modelled as:

$$C_\ell^{\rm corr}(\nu_i,\nu_j) = \rho\sqrt{A_s A_d}\left[S_s(\nu_i)\,S_d(\nu_j) + S_s(\nu_j)\,S_d(\nu_i)\right]\left(\frac{\ell}{\ell_{\rm ref}}\right)^{(\alpha_s+\alpha_d)/2}, \tag{16}$$

where $S_s(\nu) = (\nu/\nu_{\rm ref})^{\beta_s}$ is the synchrotron frequency scaling, and $\rho \in [-1,1]$ is the synchrotron-dust correlation coefficient. When color corrections are applied, each per-band scaling factor is divided by its corresponding color correction polynomial, as described in Section 4.3.4.

4.5.4 CMB subtraction

Before fitting the foreground model, the CMB contribution is subtracted from the observed power spectra. We use the best-fit theoretical ΛCDM power spectrum from Planck 2018 (Planck Collaboration, 2020), obtained from the combined Plik TTTEEE likelihood with low-multipole (lowE) and gravitational lensing constraints. Both C_ℓ^{EE} and C_ℓ^{BB} are subtracted; for BB only the lensing contribution is included, since primordial tensor B-modes have not yet been detected. The CMB spectrum is interpolated to the effective multipoles of the binning scheme before subtraction.

4.6. Bayesian inference and MCMC fitting

We adopt a Bayesian framework to constrain the foreground model parameters from the measured power spectra. The full parameter vector for the power-law fitting is $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$, where the EE and BB modes may be fitted separately or jointly. The posterior distribution is:

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{17}$$

where $\mathcal{L}$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior.

4.6.1 Likelihood

The likelihood is constructed assuming that residuals between the measured and model power spectra are Gaussian distributed. The data vector $\mathbf{D}$ consists of the measured spectra $\hat{C}_\ell(\nu_i, \nu_j)$ concatenated over all frequency pairs and multipole bins, with corresponding uncertainties $\sigma_\ell(\nu_i, \nu_j)$:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \tag{18}$$

For QUIJOTE-MFI band pairs involving channels within the same horn (11+13 GHz and 17+19 GHz), the noise is partially correlated between the two channels. In this case, a block-diagonal covariance matrix is constructed from the ensemble of 100 sky-plus-noise simulations: the off-diagonal covariance blocks for the QUIJOTE intra-horn pairs are estimated from the simulation scatter, while all remaining entries are treated as diagonal ($1/\sigma^2$). The covariance is estimated in the corrected domain (beam-deconvolved, $\mathrm{K}_{\mathrm{RJ}}^2$ units) to ensure consistency with the MCMC model. This block-diagonal inverse covariance matrix replaces the diagonal weighting for QUIJOTE correlated entries in the likelihood.

4.6.2 Priors

For the power-law fitting, uniform (flat) priors are adopted within physically motivated bounds:

$$A_s > 0, \quad A_d > 0, \quad \rho \in [-1,\, 1],$$

with broad flat ranges for the spectral indices and no additional constraint on amplitudes or slopes.

For the bin-to-bin analysis, a Gaussian prior on the synchrotron frequency spectral index regularises the fit in multipole bins where the synchrotron signal is weak:

$$\beta_s \sim \mathcal{N}(-3.1,\ 0.30^2), \tag{19}$$

consistent with published WMAP and Planck results (Planck Collaboration et al., 2016a). All remaining parameters retain flat priors.

4.6.3 Sampler

Posterior sampling is performed with the EMCEE affine-invariant ensemble sampler (Foreman-Mackey et al., 2013), implemented in the EMCEE PYTHON package. For the power-law fits, $N_{\rm walkers} = 200$ walkers are evolved for $N_{\rm iter} = 25\,000$ iterations, discarding the first 50% as burn-in. For the bin-to-bin fits, $N_{\rm walkers} = 100$ walkers are run for $N_{\rm iter} = 15\,000$ iterations with the same burn-in fraction. Convergence is assessed through visual inspection of the chain traces and by verifying that the integrated autocorrelation time is small compared to the effective chain length. Posterior summaries are reported as the median and the 16th/84th percentile credible intervals.

4.7. Fitting approaches

Two complementary fitting approaches are applied to the corrected power spectra over the range $20 \leq \ell \leq 199$, using data from all 14 frequency bands (QUIJOTE-MFI 11 and 17 GHz; WMAP 23, 33, 41, 61, and 94 GHz; Planck 30, 44, 70, 100, 143, 217, and 353 GHz). Both auto- and cross-spectra are included, yielding 105 independent spectra.

Power-law fitting. In this approach all nine multipole bins are fitted simultaneously by the global power-law foreground model of Section 4.5.. The MCMC samples the full parameter vector $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$ over all frequencies and multipoles. Three fitting configurations are explored: EE only, BB only, and a joint EE+BB analysis in which the frequency spectral indices (β_s, β_d) and the correlation coefficient (ρ) are shared between the two polarisation modes, while the amplitudes (A_s, A_d) and multipole spectral indices (α_s, α_d) are fitted independently per mode.

Bin-to-bin analysis. In this approach each multipole bin is fitted independently, without imposing any global power-law dependence on ℓ. The per-bin parameter vector is $\boldsymbol{\theta}_{\rm bin} = \{A_s, \beta_s, A_d, \beta_d, \rho\}$; the multipole spectral indices (α_s, α_d) are unconstrained in a single-bin analysis. This method allows the identification of scale-dependent behaviour and local deviations from the global power-law model assumption. Results are presented as the evolution of fitted parameters as a function of effective multipole $\ell_{\rm eff}$.

5 RESULTS

In the following subsections we first present the results obtained on simulated data and then the results for observational data, applying the three procedures described previously (Section 4.7.).

5.1. Power-law fitting

Table 1: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization auto- and cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 10°$ Galactic mask.

Data	Mode	A_s [$10^{-3}\,\mu$K^2]	α_s	β_s	A_d [$10^{-3}\,\mu$K^2]	α_d	β_d	ρ	$\chi^2_{\rm red}$
WP+Pl	EE	$6.687^{+0.421}_{-0.411}$	$-3.710^{+0.063}_{-0.065}$	$-3.017^{+0.041}_{-0.042}$	$3.217^{+0.011}_{-0.011}$	$-2.546^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.094^{+0.007}_{-0.007}$	10.15
WP+Pl	BB	$1.895^{+0.342}_{-0.317}$	$-3.258^{+0.172}_{-0.187}$	$-3.195^{+0.140}_{-0.145}$	$2.359^{+0.007}_{-0.007}$	$-2.203^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.132^{+0.013}_{-0.013}$	2.91
QJ+WP+Pl	EE	$7.398^{+0.260}_{-0.256}$	$-3.619^{+0.037}_{-0.038}$	$-3.085^{+0.016}_{-0.016}$	$3.216^{+0.011}_{-0.011}$	$-2.547^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.096^{+0.005}_{-0.005}$	7.90
QJ+WP+Pl	BB	$1.657^{+0.186}_{-0.180}$	$-3.334^{+0.112}_{-0.121}$	$-2.999^{+0.052}_{-0.052}$	$2.359^{+0.007}_{-0.007}$	$-2.203^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.111^{+0.010}_{-0.010}$	2.50

5.2. Bin-to-bin fitting

Table 2: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization auto- and cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 15°$ Galactic mask.

Data	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$5.395^{+0.425}_{-0.415}$	$-3.874^{+0.078}_{-0.082}$	$-3.042^{+0.047}_{-0.048}$	$2.093^{+0.010}_{-0.010}$	$-2.534^{+0.006}_{-0.006}$	$1.552^{+0.007}_{-0.007}$	$0.080^{+0.008}_{-0.008}$	8.72
WP+Pl	BB	$1.723^{+0.376}_{-0.342}$	$-3.284^{+0.207}_{-0.229}$	$-3.314^{+0.175}_{-0.185}$	$1.378^{+0.006}_{-0.006}$	$-2.317^{+0.006}_{-0.006}$	$1.530^{+0.007}_{-0.007}$	$0.128^{+0.016}_{-0.015}$	1.95
QJ+WP+Pl	EE	$5.990^{+0.260}_{-0.255}$	$-3.767^{+0.045}_{-0.046}$	$-3.079^{+0.017}_{-0.017}$	$2.092^{+0.010}_{-0.009}$	$-2.535^{+0.006}_{-0.006}$	$1.551^{+0.007}_{-0.007}$	$0.068^{+0.006}_{-0.006}$	6.85
QJ+WP+Pl	BB	$1.367^{+0.198}_{-0.183}$	$-3.410^{+0.141}_{-0.150}$	$-3.017^{+0.061}_{-0.061}$	$1.379^{+0.006}_{-0.006}$	$-2.317^{+0.006}_{-0.006}$	$1.531^{+0.007}_{-0.007}$	$0.116^{+0.013}_{-0.012}$	1.84

Table 3: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization auto- and cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 20°$ Galactic mask.

Data	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$5.948^{+0.451}_{-0.439}$	$-3.741^{+0.076}_{-0.080}$	$-3.140^{+0.054}_{-0.055}$	$1.439^{+0.009}_{-0.009}$	$-2.479^{+0.008}_{-0.008}$	$1.487^{+0.008}_{-0.008}$	$0.030^{+0.009}_{-0.009}$	3.36
WP+Pl	BB	$1.436^{+0.374}_{-0.336}$	$-3.476^{+0.243}_{-0.272}$	$-3.456^{+0.186}_{-0.198}$	$0.927^{+0.005}_{-0.005}$	$-2.237^{+0.007}_{-0.007}$	$1.532^{+0.009}_{-0.009}$	$0.162^{+0.019}_{-0.018}$	1.70
QJ+WP+Pl	EE	$5.880^{+0.258}_{-0.253}$	$-3.732^{+0.046}_{-0.047}$	$-3.113^{+0.019}_{-0.019}$	$1.439^{+0.009}_{-0.009}$	$-2.478^{+0.008}_{-0.008}$	$1.486^{+0.008}_{-0.008}$	$0.023^{+0.007}_{-0.007}$	2.85
QJ+WP+Pl	BB	$1.347^{+0.208}_{-0.198}$	$-3.371^{+0.151}_{-0.164}$	$-2.984^{+0.065}_{-0.065}$	$0.927^{+0.005}_{-0.005}$	$-2.238^{+0.007}_{-0.007}$	$1.533^{+0.009}_{-0.009}$	$0.121^{+0.015}_{-0.015}$	1.60

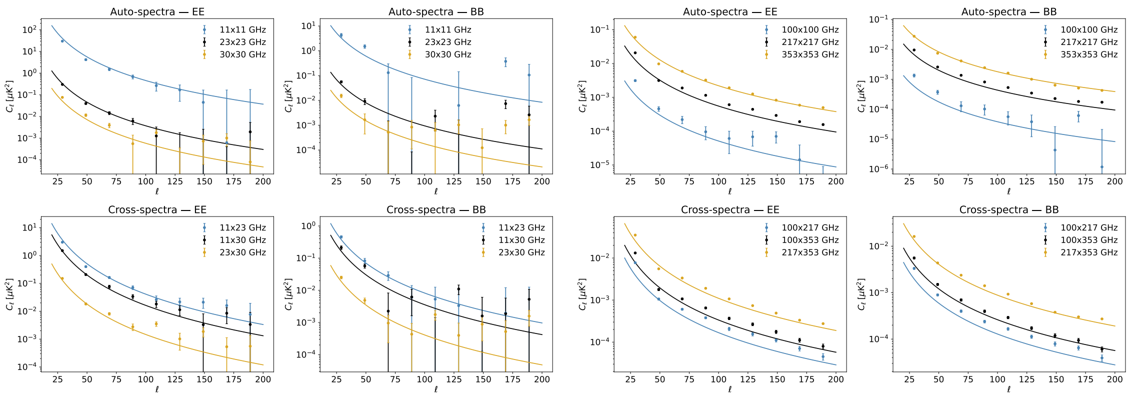

Figure 2: APS SYNCH AND DUST FIT GALCUT 10

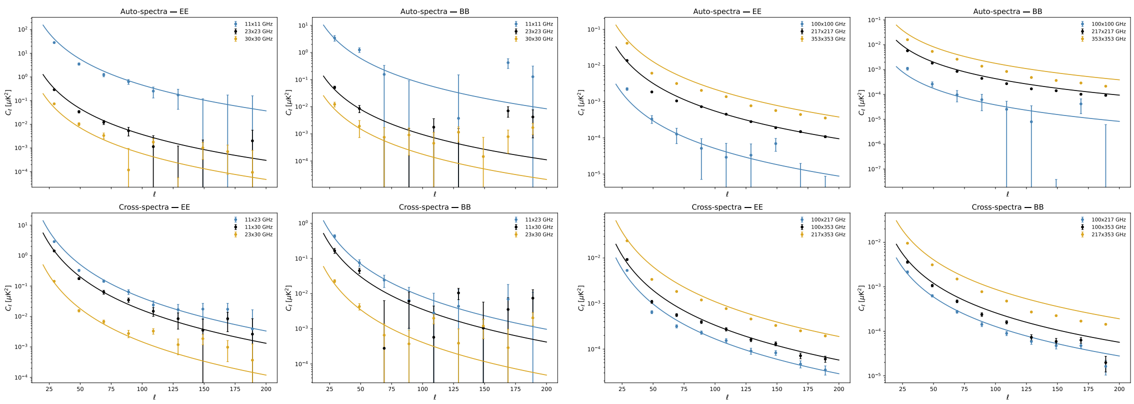

Figure 3: APS SYNCH AND DUST FIT GALCUT 15

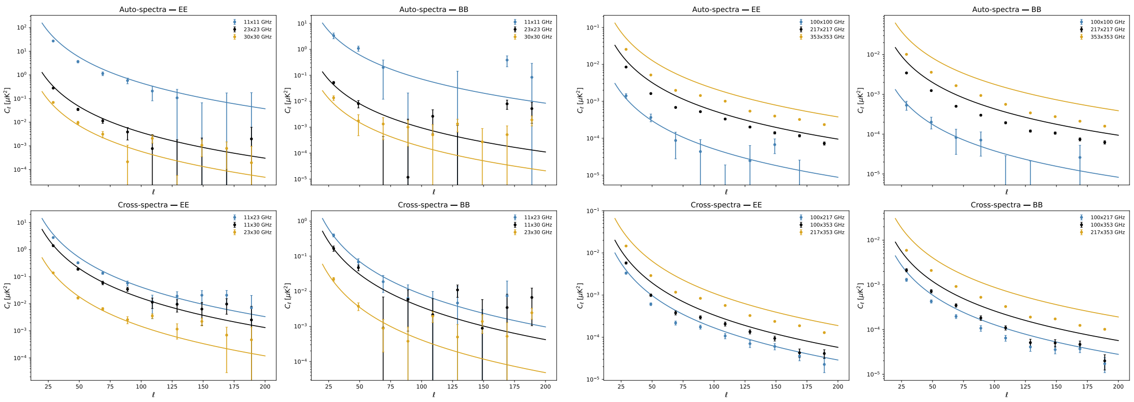

Figure 4: APS SYNCH AND DUST FIT GALCUT 20

Table 4: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	$\ell_{\rm eff}$	$A^{\rm EE}_{\rm sync}$ [10^{-3} μK^2]	$\beta^{\rm EE}_{\rm sync}$	$A^{\rm EE}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm EE}_{\rm dust}$	$\rho^{\rm EE}$	χ^2_ν
20–39	29.0	331.83 ± 3.72	-3.09 ± 0.02	75.38 ± 0.46	1.59 ± 0.01	0.092 ± 0.007	4.195
40–59	49.0	42.42 ± 1.32	-3.15 ± 0.04	12.25 ± 0.13	1.63 ± 0.01	0.117 ± 0.012	1.644
60–79	69.0	14.79 ± 0.83	-3.24 ± 0.07	7.36 ± 0.07	1.62 ± 0.01	0.046 ± 0.014	1.195
80–99	89.0	7.00 ± 0.91	-3.15 ± 0.33	4.10 ± 0.17	1.59 ± 0.09	0.050 ± 0.045	2.756
100–119	109.0	3.95 ± 0.54	-2.99 ± 0.15	2.40 ± 0.03	1.66 ± 0.02	0.030 ± 0.028	1.418
120–139	129.0	2.72 ± 0.56	-2.67 ± 0.34	1.54 ± 0.09	1.51 ± 0.15	0.168 ± 0.041	1.543
140–159	149.0	2.08 ± 0.62	-2.96 ± 0.37	1.04 ± 0.09	1.57 ± 0.18	0.230 ± 0.062	1.357
160–179	169.0	1.50 ± 0.54	-3.11 ± 0.27	0.74 ± 0.01	1.63 ± 0.03	0.065 ± 0.066	1.130
180–199	189.0	1.21 ± 0.67	-2.92 ± 0.30	0.63 ± 0.01	1.71 ± 0.03	0.108 ± 0.124	1.427
ℓ range	$\ell_{\rm eff}$	$A^{\rm BB}_{\rm sync}$ [10^{-3} μK^2]	$\beta^{\rm BB}_{\rm sync}$	$A^{\rm BB}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm BB}_{\rm dust}$	$\rho^{\rm BB}$	χ^2_ν
20–39	29.0	53.71 ± 2.06	-3.03 ± 0.06	35.01 ± 0.31	1.62 ± 0.01	0.115 ± 0.014	3.280
40–59	49.0	10.76 ± 1.80	-3.14 ± 0.48	9.39 ± 0.11	1.59 ± 0.08	0.108 ± 0.071	2.371
60–79	69.0	1.31 ± 0.55	-3.18 ± 0.25	5.17 ± 0.04	1.64 ± 0.01	0.086 ± 0.070	2.471
80–99	89.0	0.81 ± 0.46	-3.03 ± 0.23	3.09 ± 0.03	1.63 ± 0.01	0.142 ± 0.126	2.167
100–119	109.0	0.87 ± 0.46	-2.73 ± 0.25	2.05 ± 0.02	1.66 ± 0.01	0.100 ± 0.090	2.123
120–139	129.0	1.26 ± 0.45	-3.02 ± 0.23	1.27 ± 0.01	1.62 ± 0.02	0.139 ± 0.064	2.349
140–159	149.0	0.19 ± 0.31	-2.97 ± 0.31	0.81 ± 0.01	1.60 ± 0.02	-0.025 ± 0.212	1.668
160–179	169.0	1.15 ± 0.52	-3.08 ± 0.43	0.64 ± 0.02	1.62 ± 0.16	-0.059 ± 0.085	1.956
180–199	189.0	1.21 ± 0.63	-3.16 ± 0.34	0.56 ± 0.01	1.57 ± 0.02	0.059 ± 0.104	2.291

Table 5: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	$\ell_{\rm eff}$	$A_{\rm sync}^{\rm EE}$ [10^{-3} μK^2]	$\beta_{\rm sync}^{\rm EE}$	$A_{\rm dust}^{\rm EE}$ [10^{-3} μK^2]	$\beta_{\rm dust}^{\rm EE}$	$\rho^{\rm EE}$	χ^2_ν
20–39	29.0	309.76 ± 3.76	-3.09 ± 0.02	51.34 ± 0.42	1.57 ± 0.01	0.078 ± 0.008	2.418
40–59	49.0	36.06 ± 1.32	-3.12 ± 0.05	7.74 ± 0.12	1.70 ± 0.02	0.045 ± 0.015	1.358
60–79	69.0	12.41 ± 1.48	-3.24 ± 0.33	4.01 ± 0.09	1.60 ± 0.03	-0.005 ± 0.056	1.120
80–99	89.0	6.50 ± 0.65	-3.17 ± 0.12	2.61 ± 0.04	1.61 ± 0.02	0.075 ± 0.022	2.193
100–119	109.0	3.59 ± 0.53	-2.93 ± 0.17	1.72 ± 0.03	1.64 ± 0.02	0.091 ± 0.029	1.643
120–139	129.0	2.31 ± 0.56	-2.87 ± 0.37	0.94 ± 0.06	1.47 ± 0.11	0.118 ± 0.047	1.335
140–159	149.0	2.11 ± 0.65	-2.85 ± 0.27	0.71 ± 0.02	1.58 ± 0.03	0.149 ± 0.063	1.267
160–179	169.0	1.73 ± 0.63	-2.98 ± 0.50	0.56 ± 0.01	1.66 ± 0.19	0.025 ± 0.065	1.037
180–199	189.0	0.94 ± 0.66	-2.92 ± 0.31	0.45 ± 0.01	1.73 ± 0.04	0.097 ± 0.154	1.324
ℓ range	$\ell_{\rm eff}$	$A_{\rm sync}^{\rm BB}$ [10^{-3} μK^2]	$\beta_{\rm sync}^{\rm BB}$	$A_{\rm dust}^{\rm BB}$ [10^{-3} μK^2]	$\beta_{\rm dust}^{\rm BB}$	$\rho^{\rm BB}$	χ^2_ν
20–39	29.0	46.57 ± 2.16	-3.10 ± 0.07	20.32 ± 0.25	1.60 ± 0.01	0.163 ± 0.017	2.787
40–59	49.0	9.71 ± 0.84	-3.11 ± 0.12	6.76 ± 0.08	1.59 ± 0.01	0.047 ± 0.022	1.992
60–79	69.0	0.95 ± 0.54	-3.13 ± 0.26	3.30 ± 0.03	1.66 ± 0.01	0.118 ± 0.121	2.143
80–99	89.0	0.93 ± 0.50	-2.86 ± 0.25	1.72 ± 0.02	1.65 ± 0.02	0.113 ± 0.119	1.472
100–119	109.0	1.02 ± 0.53	-2.77 ± 0.39	1.07 ± 0.07	1.66 ± 0.24	0.038 ± 0.091	1.431
120–139	129.0	1.18 ± 0.47	-3.05 ± 0.26	0.61 ± 0.01	1.64 ± 0.03	0.090 ± 0.070	1.725
140–159	149.0	0.20 ± 0.32	-2.98 ± 0.31	0.47 ± 0.01	1.61 ± 0.03	0.030 ± 0.235	1.551
160–179	169.0	1.17 ± 0.50	-3.24 ± 0.28	0.36 ± 0.01	1.59 ± 0.04	-0.101 ± 0.078	1.465
180–199	189.0	1.97 ± 0.69	-2.87 ± 0.27	0.29 ± 0.01	1.60 ± 0.04	-0.011 ± 0.069	1.596

Table 6: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	$\ell_{\rm eff}$	$A_{\rm sync}^{\rm EE}$ [10^{-3} μK^2]	$\beta_{\rm sync}^{\rm EE}$	$A_{\rm dust}^{\rm EE}$ [10^{-3} μK^2]	$\beta_{\rm dust}^{\rm EE}$	$\rho^{\rm EE}$	χ^2_ν
20–39	29.0	296.73 ± 4.12	-3.12 ± 0.02	31.74 ± 0.39	1.59 ± 0.01	0.010 ± 0.010	1.857
40–59	49.0	36.01 ± 1.36	-3.13 ± 0.05	6.38 ± 0.11	1.61 ± 0.02	0.066 ± 0.015	1.634
60–79	69.0	11.78 ± 0.79	-3.22 ± 0.09	2.47 ± 0.05	1.53 ± 0.02	-0.090 ± 0.021	1.011
80–99	89.0	6.01 ± 0.66	-3.15 ± 0.13	1.80 ± 0.03	1.57 ± 0.02	0.058 ± 0.025	1.650
100–119	109.0	2.77 ± 0.55	-2.97 ± 0.20	1.26 ± 0.03	1.66 ± 0.03	0.037 ± 0.037	1.411
120–139	129.0	2.12 ± 0.61	-2.82 ± 0.46	0.65 ± 0.02	1.40 ± 0.21	0.179 ± 0.064	1.242
140–159	149.0	2.55 ± 0.62	-2.70 ± 0.23	0.50 ± 0.02	1.55 ± 0.04	0.100 ± 0.053	1.208
160–179	169.0	1.98 ± 0.62	-3.02 ± 0.28	0.41 ± 0.01	1.63 ± 0.04	0.021 ± 0.066	1.092
180–199	189.0	0.92 ± 0.63	-2.91 ± 0.29	0.29 ± 0.01	1.70 ± 0.07	0.117 ± 0.156	1.130
ℓ range	$\ell_{\rm eff}$	$A_{\rm sync}^{\rm BB}$ [10^{-3} μK^2]	$\beta_{\rm sync}^{\rm BB}$	$A_{\rm dust}^{\rm BB}$ [10^{-3} μK^2]	$\beta_{\rm dust}^{\rm BB}$	$\rho^{\rm BB}$	χ^2_ν
20–39	29.0	44.54 ± 2.21	-3.05 ± 0.07	12.83 ± 0.20	1.61 ± 0.02	0.168 ± 0.020	2.294
40–59	49.0	9.09 ± 0.89	-3.11 ± 0.13	4.50 ± 0.06	1.57 ± 0.02	0.039 ± 0.026	1.626
60–79	69.0	0.99 ± 0.56	-3.01 ± 0.27	2.08 ± 0.03	1.69 ± 0.02	0.186 ± 0.139	1.849
80–99	89.0	0.82 ± 0.51	-2.90 ± 0.26	1.18 ± 0.02	1.67 ± 0.03	0.070 ± 0.114	1.805
100–119	109.0	1.26 ± 0.59	-2.78 ± 0.39	0.71 ± 0.06	1.61 ± 0.19	0.041 ± 0.099	1.241
120–139	129.0	1.17 ± 0.49	-3.05 ± 0.27	0.43 ± 0.01	1.66 ± 0.04	0.105 ± 0.088	1.536
140–159	149.0	0.27 ± 0.40	-2.96 ± 0.31	0.36 ± 0.01	1.64 ± 0.04	-0.000 ± 0.219	1.536
160–179	169.0	1.43 ± 0.57	-3.16 ± 0.27	0.26 ± 0.01	1.61 ± 0.05	-0.086 ± 0.078	1.242
180–199	189.0	2.19 ± 0.72	-2.76 ± 0.24	0.21 ± 0.01	1.61 ± 0.06	-0.083 ± 0.071	1.317

6 CONCLUSIONS

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A Analysis fitting only cross-spectra

1.1. Power-law fitting

Table 7: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 10°$ Galactic mask.

Data	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$7.196^{+0.572}_{-0.547}$	$-3.716^{+0.076}_{-0.079}$	$-3.177^{+0.065}_{-0.067}$	$3.178^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.520^{+0.008}_{-0.008}$	$0.095^{+0.007}_{-0.007}$	7.48
WP+Pl	BB	$2.273^{+0.494}_{-0.437}$	$-3.117^{+0.191}_{-0.202}$	$-3.296^{+0.208}_{-0.232}$	$2.394^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.555^{+0.008}_{-0.008}$	$0.132^{+0.013}_{-0.013}$	2.28
QJ+WP+Pl	EE	$7.418^{+0.294}_{-0.287}$	$-3.623^{+0.042}_{-0.042}$	$-3.088^{+0.019}_{-0.020}$	$3.181^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.521^{+0.008}_{-0.007}$	$0.095^{+0.005}_{-0.005}$	5.83
QJ+WP+Pl	BB	$1.703^{+0.206}_{-0.200}$	$-3.291^{+0.122}_{-0.131}$	$-2.960^{+0.061}_{-0.061}$	$2.395^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.556^{+0.008}_{-0.008}$	$0.114^{+0.011}_{-0.011}$	2.01

Table 8: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 15°$ Galactic mask.

Data	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$5.727^{+0.580}_{-0.549}$	$-3.875^{+0.095}_{-0.100}$	$-3.182^{+0.073}_{-0.076}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.535^{+0.011}_{-0.011}$	$0.081^{+0.008}_{-0.008}$	6.46
WP+Pl	BB	$2.163^{+0.568}_{-0.498}$	$-3.180^{+0.228}_{-0.254}$	$-3.585^{+0.286}_{-0.327}$	$1.379^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.539^{+0.012}_{-0.011}$	$0.123^{+0.015}_{-0.015}$	1.63
QJ+WP+Pl	EE	$5.998^{+0.294}_{-0.287}$	$-3.764^{+0.051}_{-0.052}$	$-3.086^{+0.021}_{-0.022}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.534^{+0.011}_{-0.011}$	$0.068^{+0.006}_{-0.007}$	5.10
QJ+WP+Pl	BB	$1.363^{+0.218}_{-0.208}$	$-3.405^{+0.157}_{-0.171}$	$-3.003^{+0.072}_{-0.074}$	$1.380^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.541^{+0.011}_{-0.011}$	$0.118^{+0.013}_{-0.013}$	1.59

Table 9: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck dataset over the multipole range $20 \leq \ell \leq 199$, using the $|b| > 20°$ Galactic mask.

Data	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$6.864^{+0.654}_{-0.628}$	$-3.717^{+0.092}_{-0.098}$	$-3.379^{+0.091}_{-0.096}$	$1.410^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.460^{+0.013}_{-0.013}$	$0.032^{+0.009}_{-0.009}$	2.64
WP+Pl	BB	$1.805^{+0.584}_{-0.492}$	$-3.421^{+0.275}_{-0.310}$	$-3.885^{+0.331}_{-0.376}$	$0.925^{+0.010}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.534^{+0.016}_{-0.016}$	$0.154^{+0.019}_{-0.018}$	1.56
QJ+WP+Pl	EE	$5.908^{+0.292}_{-0.286}$	$-3.726^{+0.052}_{-0.053}$	$-3.129^{+0.024}_{-0.024}$	$1.411^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.461^{+0.013}_{-0.013}$	$0.022^{+0.007}_{-0.007}$	2.32
QJ+WP+Pl	BB	$1.358^{+0.235}_{-0.220}$	$-3.326^{+0.170}_{-0.183}$	$-2.978^{+0.075}_{-0.077}$	$0.926^{+0.011}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.539^{+0.016}_{-0.016}$	$0.123^{+0.015}_{-0.015}$	1.49

1.2. Bin-to-bin fitting

Table 10: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, frequency spectral indices, and synchrotron-dust correlation coefficient ρ, fitted independently per multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck data set, using the $|b| > 10°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$296.777^{+3.753}_{-3.784}$	$-3.078^{+0.022}_{-0.022}$	$48.896^{+0.490}_{-0.485}$	$1.477^{+0.011}_{-0.011}$	$0.091^{+0.007}_{-0.007}$	1.328
40–59	49.0	$37.435^{+1.334}_{-1.339}$	$-3.148^{+0.057}_{-0.057}$	$7.813^{+0.126}_{-0.125}$	$1.507^{+0.019}_{-0.019}$	$0.117^{+0.013}_{-0.012}$	1.186
60–79	69.0	$12.839^{+0.855}_{-0.888}$	$-3.304^{+0.095}_{-0.096}$	$4.820^{+0.080}_{-0.080}$	$1.537^{+0.020}_{-0.019}$	$0.046^{+0.014}_{-0.014}$	1.072
80–99	89.0	$6.384^{+0.635}_{-0.652}$	$-3.126^{+0.133}_{-0.135}$	$2.703^{+0.041}_{-0.040}$	$1.525^{+0.021}_{-0.020}$	$0.055^{+0.020}_{-0.019}$	1.476
100–119	109.0	$3.714^{+0.531}_{-0.554}$	$-3.013^{+0.179}_{-0.185}$	$1.529^{+0.033}_{-0.033}$	$1.550^{+0.031}_{-0.030}$	$0.033^{+0.026}_{-0.026}$	0.968
120–139	129.0	$2.833^{+0.514}_{-0.534}$	$-2.582^{+0.216}_{-0.215}$	$0.977^{+0.024}_{-0.023}$	$1.368^{+0.034}_{-0.032}$	$0.156^{+0.040}_{-0.038}$	1.457
140–159	149.0	$2.106^{+0.607}_{-0.574}$	$-2.974^{+0.237}_{-0.248}$	$0.677^{+0.017}_{-0.016}$	$1.472^{+0.039}_{-0.038}$	$0.218^{+0.052}_{-0.044}$	1.037
160–179	169.0	$1.386^{+0.527}_{-0.522}$	$-3.168^{+0.298}_{-0.283}$	$0.472^{+0.017}_{-0.016}$	$1.522^{+0.050}_{-0.047}$	$0.072^{+0.065}_{-0.056}$	1.036
180–199	189.0	$1.295^{+0.689}_{-0.671}$	$-2.902^{+0.294}_{-0.306}$	$0.413^{+0.017}_{-0.016}$	$1.685^{+0.062}_{-0.062}$	$0.113^{+0.086}_{-0.063}$	1.236
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$46.872^{+2.152}_{-2.089}$	$-3.014^{+0.069}_{-0.070}$	$22.853^{+0.315}_{-0.301}$	$1.526^{+0.017}_{-0.016}$	$0.120^{+0.014}_{-0.014}$	1.340
40–59	49.0	$9.530^{+0.800}_{-0.804}$	$-2.969^{+0.122}_{-0.123}$	$6.130^{+0.094}_{-0.092}$	$1.503^{+0.019}_{-0.019}$	$0.111^{+0.022}_{-0.021}$	1.318
60–79	69.0	$1.411^{+0.534}_{-0.535}$	$-3.154^{+0.254}_{-0.258}$	$3.440^{+0.052}_{-0.052}$	$1.573^{+0.020}_{-0.020}$	$0.083^{+0.047}_{-0.038}$	1.616
80–99	89.0	$0.935^{+0.453}_{-0.451}$	$-3.013^{+0.232}_{-0.238}$	$2.048^{+0.031}_{-0.032}$	$1.584^{+0.023}_{-0.022}$	$0.139^{+0.072}_{-0.049}$	1.429
100–119	109.0	$0.792^{+0.476}_{-0.468}$	$-2.733^{+0.251}_{-0.270}$	$1.355^{+0.024}_{-0.023}$	$1.605^{+0.025}_{-0.025}$	$0.113^{+0.084}_{-0.051}$	1.217
120–139	129.0	$1.176^{+0.427}_{-0.424}$	$-3.019^{+0.246}_{-0.253}$	$0.832^{+0.016}_{-0.015}$	$1.559^{+0.029}_{-0.028}$	$0.137^{+0.055}_{-0.043}$	1.607
140–159	149.0	$0.320^{+0.495}_{-0.268}$	$-2.952^{+0.317}_{-0.314}$	$0.541^{+0.014}_{-0.014}$	$1.555^{+0.039}_{-0.038}$	$-0.016^{+0.098}_{-0.104}$	1.156
160–179	169.0	$0.536^{+0.505}_{-0.413}$	$-3.063^{+0.308}_{-0.308}$	$0.439^{+0.013}_{-0.013}$	$1.606^{+0.045}_{-0.044}$	$-0.064^{+0.074}_{-0.103}$	1.410
180–199	189.0	$0.794^{+0.586}_{-0.506}$	$-3.283^{+0.307}_{-0.296}$	$0.404^{+0.011}_{-0.010}$	$1.651^{+0.043}_{-0.043}$	$0.103^{+0.097}_{-0.064}$	1.375

Table 11: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, frequency spectral indices, and synchrotron-dust correlation coefficient ρ, fitted independently per multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck data set, using the $|b| > 15°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$276.030^{+3.804}_{-3.920}$	$-3.088^{+0.023}_{-0.023}$	$32.899^{+0.460}_{-0.452}$	$1.443^{+0.016}_{-0.016}$	$0.077^{+0.008}_{-0.008}$	1.573
40–59	49.0	$31.665^{+1.343}_{-1.301}$	$-3.115^{+0.068}_{-0.069}$	$4.864^{+0.121}_{-0.118}$	$1.565^{+0.030}_{-0.030}$	$0.046^{+0.015}_{-0.015}$	0.975
60–79	69.0	$10.732^{+0.864}_{-0.867}$	$-3.327^{+0.111}_{-0.115}$	$2.644^{+0.060}_{-0.059}$	$1.529^{+0.030}_{-0.028}$	$-0.004^{+0.018}_{-0.018}$	1.077
80–99	89.0	$6.012^{+0.646}_{-0.648}$	$-3.153^{+0.137}_{-0.143}$	$1.708^{+0.036}_{-0.035}$	$1.541^{+0.029}_{-0.028}$	$0.079^{+0.021}_{-0.021}$	1.461
100–119	109.0	$3.355^{+0.532}_{-0.520}$	$-2.925^{+0.179}_{-0.186}$	$1.096^{+0.029}_{-0.028}$	$1.530^{+0.037}_{-0.037}$	$0.093^{+0.029}_{-0.028}$	1.239
120–139	129.0	$2.466^{+0.557}_{-0.563}$	$-2.738^{+0.246}_{-0.250}$	$0.600^{+0.019}_{-0.019}$	$1.333^{+0.043}_{-0.041}$	$0.109^{+0.045}_{-0.042}$	1.401
140–159	149.0	$2.033^{+0.648}_{-0.631}$	$-2.921^{+0.275}_{-0.280}$	$0.450^{+0.014}_{-0.014}$	$1.443^{+0.051}_{-0.048}$	$0.142^{+0.054}_{-0.049}$	1.114
160–179	169.0	$1.723^{+0.578}_{-0.562}$	$-3.042^{+0.271}_{-0.282}$	$0.369^{+0.016}_{-0.015}$	$1.598^{+0.064}_{-0.061}$	$0.035^{+0.058}_{-0.055}$	0.983
180–199	189.0	$0.969^{+0.733}_{-0.658}$	$-2.901^{+0.307}_{-0.307}$	$0.295^{+0.016}_{-0.015}$	$1.716^{+0.090}_{-0.086}$	$0.103^{+0.119}_{-0.078}$	1.239
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu{\rm K}^2$]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$40.394^{+2.272}_{-2.243}$	$-3.130^{+0.083}_{-0.085}$	$13.261^{+0.253}_{-0.252}$	$1.497^{+0.022}_{-0.022}$	$0.168^{+0.017}_{-0.017}$	1.284
40–59	49.0	$8.505^{+0.860}_{-0.893}$	$-2.987^{+0.152}_{-0.146}$	$4.382^{+0.082}_{-0.086}$	$1.492^{+0.024}_{-0.023}$	$0.045^{+0.024}_{-0.023}$	1.596
60–79	69.0	$1.059^{+0.551}_{-0.556}$	$-3.099^{+0.261}_{-0.263}$	$2.176^{+0.043}_{-0.043}$	$1.586^{+0.027}_{-0.027}$	$0.112^{+0.078}_{-0.053}$	1.476
80–99	89.0	$0.933^{+0.494}_{-0.506}$	$-2.873^{+0.246}_{-0.262}$	$1.114^{+0.027}_{-0.027}$	$1.557^{+0.035}_{-0.035}$	$0.113^{+0.086}_{-0.058}$	1.275
100–119	109.0	$1.052^{+0.498}_{-0.501}$	$-2.853^{+0.278}_{-0.288}$	$0.694^{+0.018}_{-0.018}$	$1.592^{+0.040}_{-0.039}$	$0.038^{+0.056}_{-0.049}$	1.047
120–139	129.0	$1.146^{+0.443}_{-0.439}$	$-3.103^{+0.279}_{-0.281}$	$0.391^{+0.015}_{-0.014}$	$1.567^{+0.058}_{-0.056}$	$0.082^{+0.062}_{-0.050}$	1.331
140–159	149.0	$0.377^{+0.532}_{-0.310}$	$-2.953^{+0.307}_{-0.305}$	$0.335^{+0.014}_{-0.013}$	$1.660^{+0.067}_{-0.064}$	$0.032^{+0.119}_{-0.092}$	1.116
160–179	169.0	$0.610^{+0.509}_{-0.437}$	$-3.186^{+0.292}_{-0.278}$	$0.238^{+0.012}_{-0.011}$	$1.527^{+0.076}_{-0.072}$	$-0.126^{+0.085}_{-0.137}$	1.296
180–199	189.0	$1.485^{+0.677}_{-0.689}$	$-2.987^{+0.263}_{-0.280}$	$0.225^{+0.012}_{-0.011}$	$1.786^{+0.087}_{-0.082}$	$0.029^{+0.066}_{-0.061}$	1.123

Table 12: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, frequency spectral indices, and synchrotron-dust correlation coefficient ρ, fitted independently per multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QUIJOTE+WMAP+Planck data set, using the $|b| > 20°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$264.245^{+4.366}_{-4.499}$	$-3.116^{+0.027}_{-0.027}$	$20.371^{+0.406}_{-0.402}$	$1.456^{+0.021}_{-0.022}$	$0.009^{+0.010}_{-0.010}$	1.518
40–59	49.0	$31.624^{+1.382}_{-1.419}$	$-3.125^{+0.069}_{-0.071}$	$4.020^{+0.105}_{-0.106}$	$1.472^{+0.031}_{-0.031}$	$0.066^{+0.015}_{-0.015}$	1.503
60–79	69.0	$10.313^{+0.794}_{-0.813}$	$-3.266^{+0.112}_{-0.120}$	$1.631^{+0.050}_{-0.049}$	$1.446^{+0.039}_{-0.039}$	$-0.088^{+0.021}_{-0.022}$	0.992
80–99	89.0	$5.609^{+0.661}_{-0.676}$	$-3.129^{+0.153}_{-0.156}$	$1.172^{+0.034}_{-0.033}$	$1.493^{+0.040}_{-0.038}$	$0.062^{+0.023}_{-0.024}$	1.331
100–119	109.0	$2.531^{+0.546}_{-0.545}$	$-2.962^{+0.215}_{-0.224}$	$0.812^{+0.029}_{-0.029}$	$1.556^{+0.053}_{-0.050}$	$0.039^{+0.037}_{-0.035}$	1.110
120–139	129.0	$2.210^{+0.555}_{-0.543}$	$-2.773^{+0.250}_{-0.247}$	$0.412^{+0.017}_{-0.016}$	$1.240^{+0.057}_{-0.053}$	$0.163^{+0.056}_{-0.049}$	1.305
140–159	149.0	$2.518^{+0.630}_{-0.617}$	$-2.777^{+0.236}_{-0.252}$	$0.321^{+0.014}_{-0.014}$	$1.432^{+0.067}_{-0.065}$	$0.094^{+0.050}_{-0.047}$	1.120
160–179	169.0	$1.884^{+0.598}_{-0.566}$	$-3.110^{+0.293}_{-0.289}$	$0.274^{+0.015}_{-0.014}$	$1.611^{+0.084}_{-0.080}$	$0.038^{+0.059}_{-0.058}$	1.063
180–199	189.0	$0.973^{+0.714}_{-0.648}$	$-2.899^{+0.295}_{-0.303}$	$0.191^{+0.015}_{-0.013}$	$1.666^{+0.136}_{-0.125}$	$0.117^{+0.130}_{-0.094}$	1.099
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$37.722^{+2.289}_{-2.316}$	$-3.070^{+0.085}_{-0.088}$	$8.376^{+0.211}_{-0.209}$	$1.523^{+0.032}_{-0.032}$	$0.177^{+0.021}_{-0.020}$	1.741
40–59	49.0	$7.944^{+0.884}_{-0.899}$	$-3.016^{+0.162}_{-0.160}$	$2.934^{+0.071}_{-0.068}$	$1.480^{+0.031}_{-0.029}$	$0.036^{+0.027}_{-0.027}$	1.258
60–79	69.0	$0.949^{+0.602}_{-0.586}$	$-2.977^{+0.278}_{-0.275}$	$1.318^{+0.038}_{-0.037}$	$1.568^{+0.042}_{-0.040}$	$0.174^{+0.129}_{-0.072}$	1.527
80–99	89.0	$0.906^{+0.528}_{-0.534}$	$-2.941^{+0.262}_{-0.271}$	$0.755^{+0.026}_{-0.025}$	$1.568^{+0.051}_{-0.048}$	$0.066^{+0.083}_{-0.065}$	1.670
100–119	109.0	$1.235^{+0.538}_{-0.528}$	$-2.874^{+0.283}_{-0.295}$	$0.466^{+0.016}_{-0.016}$	$1.543^{+0.052}_{-0.051}$	$0.049^{+0.060}_{-0.054}$	1.043
120–139	129.0	$1.116^{+0.468}_{-0.484}$	$-3.117^{+0.285}_{-0.279}$	$0.278^{+0.014}_{-0.014}$	$1.614^{+0.084}_{-0.082}$	$0.097^{+0.076}_{-0.060}$	1.160
140–159	149.0	$0.457^{+0.599}_{-0.375}$	$-2.938^{+0.305}_{-0.304}$	$0.272^{+0.014}_{-0.014}$	$1.794^{+0.091}_{-0.087}$	$0.017^{+0.115}_{-0.098}$	1.128
160–179	169.0	$0.840^{+0.578}_{-0.542}$	$-3.096^{+0.274}_{-0.276}$	$0.173^{+0.011}_{-0.010}$	$1.523^{+0.102}_{-0.095}$	$-0.103^{+0.080}_{-0.111}$	1.111
180–199	189.0	$1.683^{+0.697}_{-0.731}$	$-2.855^{+0.245}_{-0.271}$	$0.154^{+0.011}_{-0.010}$	$1.706^{+0.123}_{-0.113}$	$-0.054^{+0.069}_{-0.073}$	1.158